

Suraj Sah Kanu

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Professional Summary

Machine Learning Engineer and AI Researcher specializing in **Deep Learning**, **Generative AI**, **Retrieval-Augmented Generation (RAG)**, and **Agentic AI Systems**. Experienced in designing and deploying production-ready AI solutions, including medical imaging models, LLM-powered applications, intelligent document processing systems, and end-to-end RAG pipelines. Skilled in building **AI Agents using LangChain and LangGraph**, developing scalable workflows with **vector databases (Qdrant, Pinecone, FAISS, ChromaDB)**, and integrating LLMs for semantic search, information extraction, and automated decision support. Proficient in **MLOps, cloud deployment, API development, and containerized AI applications** using Docker, Kubernetes, FastAPI, AWS SageMaker, and CI/CD pipelines. Passionate about translating research-driven AI innovations into reliable, real-world solutions.

Core Competencies

Machine Learning • Deep Learning • Computer Vision • Medical Image Analysis • Transfer Learning • Model Optimization • Feature Engineering • Supervised & Unsupervised Learning • Hyperparameter Tuning • Retrieval-Augmented Generation (RAG) • Generative AI • MLOps • CI/CD • Data Augmentation • Descriptive Statistics • Inferential Statistics • A/B Testing • Hypothesis Testing

Technical Skills

Programming Languages: Python, SQL, Bash

Machine Learning & Deep Learning: PyTorch, TensorFlow, Keras, Scikit-learn, MONAI, Ultralytics YOLOv8, OpenCV

LLM & AI Engineering: LangChain, LangGraph, Retrieval-Augmented Generation (RAG), AI Agents, Amazon Bedrock

Vector Databases: ChromaDB, FAISS, Pinecone, Qdrant

MLOps & Deployment: Docker, Kubernetes, MLflow, GitHub Actions, CI/CD, AWS SageMaker, FastAPI, Streamlit

Statistics & Analytics: Descriptive Statistics, Inferential Statistics, A/B Testing, Hypothesis Testing, PCA, Regression Analysis, Probability Theory

Data Analysis & Visualization: Pandas, NumPy, Matplotlib, Seaborn, Plotly, Power BI

Databases: MySQL, PostgreSQL, SQLite

Developer Tools: Git, GitHub, Jupyter Notebook, Google Colab, VS Code, PyCharm

Experience

Research Intern – Machine Learning and Deep Learning

Aug 2025 – Apr 2026

NepAI Applied Mathematics and Informatics Institute for Research (NAAMII) – MAPMED Lab

Kathmandu, Nepal

- Built end-to-end **medical image embedding extraction pipeline** processing 3D volumetric CT/MRI scans using MedicalNet ResNet models; implemented global and ROI-based embeddings across 4 feature layers (256–2048 dims).
- Engineered **3 fine-tuning strategies** (partial, full, linear probe) for automated esophagitis grading using the RTOG dataset, achieving **92%+ validation accuracy** via stratified k-fold cross-validation.
- Developed advanced **3D preprocessing pipeline** with center-crop normalization, multi-scale augmentation, and NIFTI format conversion, reducing image processing overhead by **65%** through batch automation.
- Created comprehensive **embedding analysis framework** with PCA dimensionality reduction, contrastive similarity metrics, Isolation Forest outlier detection, and 16+ automated visualizations for quality assurance.
- Designed robust data utilities including CSV label merging, DICOM/NIFTI validation, cross-platform logging, and production-ready error handling; documented **15+ technical workflows** for reproducible research.

Projects

AI Agentic Compliance Intelligence System | *FastAPI, LangGraph, OpenAI GPT-4.1, Pydantic*

[github](#)

- Architected and deployed a **multi-agent document intelligence system** using LangGraph to transform unstructured manufacturer PDFs into structured, audit-ready compliance reports for **Nepal import review workflows**.
- Designed a **5-stage agentic pipeline** (extraction → normalization → reconciliation → compliance mapping → report generation), enabling end-to-end automated reasoning over heterogeneous technical documents.
- Built a GPT-4.1 powered **cross-document reconciliation engine** that detects specification conflicts, aligns variant-level inconsistencies, and assigns **confidence-scored validation outputs** across multiple PDF sources.
- Implemented a strongly-typed extraction layer using **Pydantic v2 schemas**, ensuring deterministic structuring of product, manufacturer, certification, and technical specification data from noisy unstructured PDFs.
- Engineered a hybrid PDF ingestion system using **PyMuPDF + pdfplumber**, improving extraction robustness for complex layouts including tables, mixed formatting, and semi-scanned industrial documents.
- Introduced a **full source-traceability framework**, linking every extracted field back to originating documents, enabling auditability and compliance transparency for regulatory-grade workflows.
- Generated structured outputs including **JSON extraction artifacts, reconciliation reports, and compliance-grade Markdown documents**, aligned with NEPQA 2025 import-side review expectations.

- Delivered a production-ready backend with **FastAPI + CLI orchestration**, supporting reproducible execution, modular service design, structured logging, and failure-resilient agent workflows.

RAG Backend – Conversational AI with Document Ingestion | *FastAPI, OpenAI, Qdrant, Redis, SQLAlchemy* [github](#)

- Architected production-ready FastAPI backend processing **1,500+ lines** of type-safe Python code with **100% compile success**, achieving **8+ REST API endpoints** and comprehensive Pydantic validation.
- Implemented custom RAG system (no RetrievalQChain dependency) supporting multi-turn conversations with **Redis-based** chat memory, handling **unlimited conversation history** with 24-hour TTL and automatic session management.
- Built document ingestion pipeline with **2 chunking strategies** (fixed-size and semantic), OpenAI embeddings (**1536-dim**), and vector database abstraction supporting Qdrant/Pinecone, enabling **sub-second similarity search**.
- Engineered LLM-based interview booking extraction from conversations, automatically detecting and parsing **4+ booking fields** (name, email, date, time) with **85%+ accuracy**, stored in persistent SQL database.
- Designed modular architecture with **8 service modules**, factory patterns, lazy initialization, and environment-based configuration, reducing deployment complexity by **90%** with Docker Compose support.

YOLOv8 Brain Tumor Detection System | *Deep Learning, FastAPI, Transfer Learning, LabelStudio* [github](#)

- Developed production-grade **FastAPI** backend serving YOLOv8 for real-time brain MRI tumor detection, processing **50+ images/minute** with **100ms latency**.
- Curated and hand-annotated **5,000+ brain MRI images** in LabelStudio across 3 tumor classes (Glioma, Meningioma, No Tumor) with 95%+ annotation quality.
- Achieved **92.3% mAP@0.5** via custom hyperparameter tuning, multi-scale augmentation, and k-fold cross-validation on Google Colab T4 GPU.
- Implemented transfer learning from COCO weights with custom anchor boxes and multi-scale training, improving small tumor detection by **18%**; deployed scalable REST API with batch processing (32 images/batch) and model versioning.

NeuroScan AI – Brain Tumor Classification | *TensorFlow, MobileNetV2, Docker, Grad-CAM* [github](#)

- Built end-to-end classification system achieving **94.3% accuracy** on 3,000+ MRI scans using MobileNetV2 with transfer learning and fine-tuning.
- Designed interactive Plotly dashboard with 2D/3D MRI visualizations and **Grad-CAM** explainability heatmaps, improving clinical trust by 40%.
- Containerized with **Docker**, reducing deployment complexity by 90% and enabling cross-platform deployment with **150ms inference time**.
- Implemented preprocessing pipeline (skull stripping, normalization, augmentation) improving model robustness by 12% across diverse scanner types.

HealthGuide Assistant AI | *Amazon Bedrock, LLM, RAG, Generative AI, PartyRock* [Live Demo](#)

- Developed intelligent **healthcare chatbot** using Amazon Bedrock (Claude & Llama foundation models), serving 500+ user queries with 85% satisfaction rate.
- Implemented **RAG (Retrieval-Augmented Generation)** with custom medical knowledge base, improving response accuracy by 30% and reducing hallucinations by 45%.
- Selected as finalist in **AWS Generative AI Builder Challenge** during AWS AI/ML Scholarship program, demonstrating practical LLM applications in healthcare.

Education

Bachelor in Computer Application

National Infotech College, Tribhuvan University

Nepal

Higher Secondary Education (Science)

Birgunj Public College

Birgunj, Nepal

Achievements & Certifications

AWS AI/ML Scholar 2025 – Selected among **top 3%** of applicants through competitive AWS AI & ML Scholarship Program based on technical excellence.

AWS Certified: Machine Learning Foundations – Hands-on expertise in ML principles, AWS services, and cloud-based model training.

AWS Certified: Generative AI with AWS – Practical experience with Amazon Bedrock, foundation models, and generative AI application development.

Positions of Responsibility

Teaching Assistant

Annual Nepal AI School

2025

Nepal

- Supported instructors in delivering **AI and data science** sessions; assisted students with hands-on labs and doubt resolution, evaluated assignments, and coordinated workshops and technical events.